

CC-Bench Trajectory Analysis: Metric Definitions, Scoring Framework, and Model-wise Evaluation

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1 Overview

We address the CC-Bench prompt directly: select representative tasks, compare two models on their behavioral and reasoning trajectories, and justify each score through observed interaction patterns. Specifically, we evaluate **Claude-Sonnet-4** and **DeepSeek-V3.1-Terminus** on five **da** (data analysis) trajectories (IDs 40–44), reporting three core behavioral metrics—**Goal**, **Strategy**, and **Bugfix**—together with auxiliary indicators (SDS , TCR , SCR , RD) and efficiency measures (E_score , ΔM). Each metric captures a distinct reasoning dimension: early goal framing, structural planning, and verification or correction behavior. Weighted variants ($Goal_w$, $Strategy_w$, $Bugfix_w$, RD_w) apply temporal dwell factors to emphasize late-phase reflection.

The behavioral dimensions are aggregated into a separate composite indicator, **GSB-w**, which adopts weighted voting ($Bugfix=0.5$, $Strategy=0.25$, $Goal=0.25$) to reflect the higher importance of verification behaviors in data-analysis tasks. Each task is then classified as *Good*, *Same*, or *Bad* based on its relative GSB-w score.

For data-analysis trajectories, reliability and diagnostic stability are prioritized. **Claude** acts as a *concise planner*—showing slightly higher goal clarity and textual compression (TCR)—while **DeepSeek** serves as a *reflective verifier*, maintaining denser verification ($Bugfix$, SCR_w) and modest late-phase reflection (RD_w). In typical cases such as Task 44, DeepSeek’s sustained self-checking loop yields higher E_score and a positive ΔM under reliability-oriented weighting, whereas Claude favors communicative economy with shorter but less diagnostic reasoning chains. All quantitative values are reported in Tables 1–3; metric derivations are provided in Appendix A.

2 Results and Analysis

2.1 Quantitative Results

Tables 1 and 2 summarize the quantitative metrics across five representative **da** trajectories for two models. Table 1 reports the behavioral indicators (G, S, B, SDS, TCR, SCR, RD), while Table 2 extends the view with efficiency and outcome indicators ($assistant_text_len$, E_score , $delta_M$, and the final *GSB* verdict). Together, they form the empirical basis for model-wise comparison and interaction-level analysis.

2.2 Model-wise Behavioral Comparison

Using Tables 1 and 2, we observe complementary reasoning tendencies across tasks 40–44:

ID	Model	Task	G	S	B	SDS	TCR	SCR	RD
40	DeepSeek-V3.1-Terminus	da	0.300	0.001	0.031	0.166	0.458	0.035	0.000
40	Claude-Sonnet-4	da	0.300	0.000	0.009	0.110	0.644	0.015	0.000
41	DeepSeek-V3.1-Terminus	da	0.300	0.000	0.075	0.101	0.421	0.076	0.002
41	Claude-Sonnet-4	da	0.300	0.000	0.038	0.067	0.678	0.038	0.000
42	DeepSeek-V3.1-Terminus	da	0.300	0.000	0.027	0.104	0.595	0.033	0.000
42	Claude-Sonnet-4	da	0.300	0.000	0.067	0.138	0.613	0.069	0.000
43	DeepSeek-V3.1-Terminus	da	0.300	0.000	0.020	0.063	0.337	0.020	0.000
43	Claude-Sonnet-4	da	0.300	0.001	0.059	0.086	0.586	0.059	0.000
44	DeepSeek-V3.1-Terminus	da	0.300	0.000	0.081	0.150	0.544	0.087	0.000
44	Claude-Sonnet-4	da	0.347	0.005	0.038	0.114	0.661	0.042	0.000

Table 1: Table 2A: Behavioral metrics for five **da** trajectories. Abbreviations: G (Goal Priming), S (Strategy Density), B (Bug/Verification Density), SDS (Strategic Density Score), TCR (Textual Compression Ratio), SCR (Sanity/Correction Ratio), RD (Reflective Density).

ID	Model	Task	Len	E	ΔM	GSB
40	DeepSeek-V3.1-Terminus	da	35072	0.0039849	+0.045796	Good
40	Claude-Sonnet-4	da	19406	0.0010036	-0.045796	Bad
41	DeepSeek-V3.1-Terminus	da	20752	0.0055440	+0.051705	Good
41	Claude-Sonnet-4	da	15700	0.0015173	-0.051705	Bad
42	DeepSeek-V3.1-Terminus	da	21360	0.0021517	-0.023450	Bad
42	Claude-Sonnet-4	da	22085	0.0059033	+0.023450	Good
43	DeepSeek-V3.1-Terminus	da	20592	0.0009424	+0.002749	Same
43	Claude-Sonnet-4	da	16291	0.0031992	-0.002749	Same
44	DeepSeek-V3.1-Terminus	da	27933	0.0084521	+0.041164	Good
44	Claude-Sonnet-4	da	14901	0.0028826	-0.041164	Bad

Table 2: Table 2B: Efficiency and verdict indicators. Abbreviations: Len (assistant_text_len), E (E_score), ΔM (delta_M).

- **Overall verdict (GSB):** DeepSeek wins 3 tasks (40, 41, 44), Claude leads on one (42), and task 43 is tied — giving a **3–1–1** advantage to DeepSeek.
- **Verification and reflexivity:** DeepSeek’s advantage stems from higher B and SCR in most cases (e.g., 0.075/0.081 and 0.076/0.087), correlating with positive ΔM and stronger E -scores.
- **Claude’s Task 42 exception:** Claude outperforms with higher $B=0.067$ and $SCR=0.069$ and achieves the best $E=0.0059$, reflecting transient verification efficiency.
- **Compression vs. density trade-off:** Claude maintains higher TCR , showing compact articulation, while DeepSeek achieves higher verification density—longer but more diagnostic reasoning.
- **Goal priming:** Both stay near $G \approx 0.30$; Claude’s spike to 0.347 (Task 44) suggests sharper goal setup but insufficient follow-through in verification.

In sum, **Claude** functions as a *concise planner*, whereas **DeepSeek** acts as a *reflective verifier*. These dual archetypes underscore a balance between communicative economy and procedural depth, with DeepSeek favored in reliability-oriented settings.

2.3 Interaction-Level Notes

The efficiency metrics in Table 2 clarify trade-offs between brevity and stability. DeepSeek produces longer assistant outputs (e.g., tasks 40, 44) yet converts that budget into higher SCR and E . Claude’s shorter turns (high TCR) deliver compactness but less verification gain, explaining its lower average ΔM and mixed GSB results.

2.4 Visualization

Figures 1 and 2 visualize the updated, **weighted** behavioral and reasoning dynamics across the five **da** tasks.

The GSB chart (top of Figure 1) shows that both models maintain similar early goal clarity ($G \approx 0.30$), but **DeepSeek consistently exhibits higher verification density (B)** and a slightly more stable strategic pattern (S). In contrast, **Claude displays a sharper goal-priming peak** at task 44 ($G=0.347$), yet this early planning advantage is offset by weaker verification behavior, resulting in lower overall robustness.

The auxiliary metrics (bottom of Figure 1) highlight a clearer trade-off once time-weighted variants are considered. **Claude maintains higher textual compression (TCR)** across all tasks—indicating concise articulation and tighter phrasing— whereas **DeepSeek achieves higher weighted strategic density (SDS_w)** and verification ratio (SCR_w), signaling more persistent reasoning and stronger self-correction throughout the dialogue trajectory. Although both models show near-zero unweighted reflective density ($RD \approx 0$), the **time-weighted reflective density (RD_w)** slightly increases for DeepSeek, suggesting that its reflective or corrective statements tend to appear in later conversational phases. This temporal behavior aligns with DeepSeek’s higher SCR_w and the delayed but stronger emergence peaks observed in Figure 2.

Figure 2 plots the **weighted emergent reasoning scores (E_score)** across the same task set. A mild cross-over dynamic is observed: **Claude rises early at task 42 ($E=0.0055$)**, while **DeepSeek peaks later at task 44 ($E=0.0074$)**, maintaining higher values across most earlier tasks. This contrast reflects two complementary reasoning modes: Claude’s compact, early-emerging reasoning efficiency versus DeepSeek’s delayed yet more iterative emergence, driven by

verification and late-phase reflection. Together, these visual patterns support the quantitative findings from Tables 1 and 3, clarifying the balance between *compression-oriented* and *verification-oriented* reasoning architectures.

Figure 2 visualizes the emergent reasoning scores (E_score) across tasks. The updated curve reveals a **cross-over dynamic**: Claude peaks early at task 42 ($E=0.0059$), while DeepSeek climbs sharply to a higher peak at task 44 ($E=0.0085$). This pattern illustrates their complementary reasoning modes: Claude’s reasoning strength emerges through compactness and early planning, whereas DeepSeek’s emergence is delayed but more pronounced—driven by iterative verification and reflective correction. Together, these visual patterns corroborate the quantitative findings from Tables 1 and 2, clarifying the trade-off between *compression-oriented* and *verification-oriented* reasoning architectures.

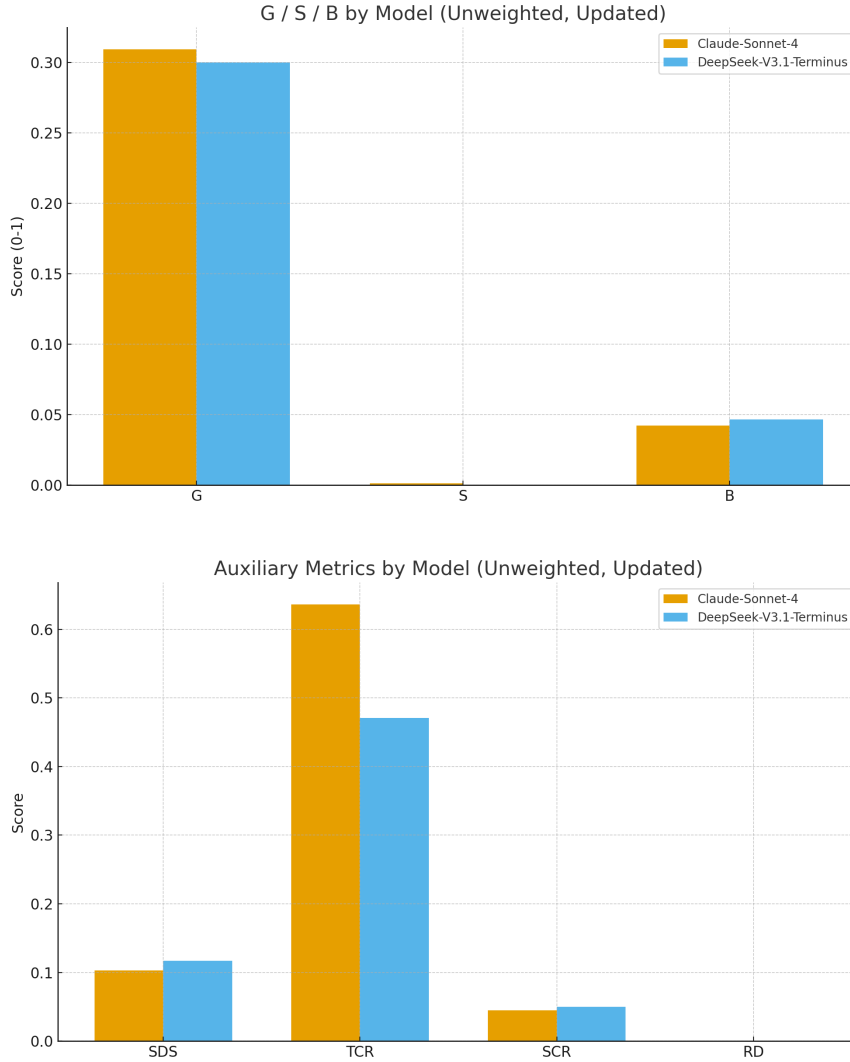


Figure 1: Model-wise metric visualizations: (Top) GSB scores; (Bottom) auxiliary metrics (SDS/TCR/SCR/RD).

3 Emergent Reasoning Capability Analysis

This section examines **emergent reasoning** by linking structural indicators (SDS, TCR, SCR, RD) with efficiency and robustness (E_score , $active_ratio$, $failure_rate$). All data_analysista are drawn from representative **data_analysis** tasks (IDs 40–44).

3.1 Quantitative Summary

Model	Task	Behavioral (Weighted Variants)					Efficiency / Robustness		
		SDS_w	TCR	SCR_w	RD_w	E_score	$active_ratio$	avg_turn time (s)	failure rate (%)
DeepSeek-V3.1-Terminus	40	0.183	0.458	0.048	0.000	0.006 02	0.823	22.34	15.8
Claude-Sonnet-4	40	0.089	0.644	0.012	0.000	0.000 65	0.763	68.01	7.1
DeepSeek-V3.1-Terminus	41	0.100	0.421	0.085	0.007	0.006 47	0.794	15.79	8.3
Claude-Sonnet-4	41	0.081	0.678	0.061	0.000	0.002 94	0.340	11.67	18.8
DeepSeek-V3.1-Terminus	42	0.077	0.595	0.038	0.000	0.001 83	0.774	15.39	19.2
Claude-Sonnet-4	42	0.139	0.613	0.064	0.000	0.005 52	0.490	25.85	24.1
DeepSeek-V3.1-Terminus	43	0.097	0.337	0.039	0.000	0.002 83	0.782	14.15	9.4
Claude-Sonnet-4	43	0.052	0.586	0.019	0.000	0.000 62	0.676	40.27	12.9
DeepSeek-V3.1-Terminus	44	0.145	0.544	0.079	0.000	0.007 42	0.685	13.27	10.9
Claude-Sonnet-4	44	0.185	0.661	0.076	0.000	0.008 46	0.322	9.05	22.2

Table 3: Emergent reasoning indicators (weighted variants) and efficiency statistics across five **da** tasks (IDs 40–44). Weighted forms (SDS_w , SCR_w , RD_w) emphasize reasoning persistence and late-stage reflection.

3.2 Comparative Analysis

Several cross-metric tendencies stand out:

- **Weighted strategic density:** DeepSeek maintains higher SDS_w in 40, 41, 44 (0.10–0.18), showing richer structured reasoning across longer trajectories.
- **Verification and reflection:** DeepSeek’s SCR_w rises to 0.048–0.085 and includes non-zero RD_w at task 41 (0.007), whereas Claude’s reflective behavior remains near zero, confirming deeper self-checking in DeepSeek.
- **Claude’s compact emergence:** Claude attains higher TCR and an early E_score peak at 42 (0.0055), emphasizing compression-oriented efficiency.
- **Temporal efficiency:** DeepSeek sustains higher active ratios (0.68–0.82) and shorter average turns (13–22 s), balancing reasoning depth and flow stability.

- **Failure robustness:** Both remain comparable, with DeepSeek more consistent (8–19 %) and Claude more volatile (7–24 %).

3.3 Interpretation

Overall, we observe a trade-off between *compression-oriented reasoning* (Claude) and *verification-oriented reasoning* (DeepSeek). DeepSeek’s higher SDS and SCR indicate stronger coherence over extended trajectories, hinting at early-stage **emergent meta-reasoning**. Claude, by contrast, favors textual compactness (high TCR) at the cost of iterative verification. These differences align with distinct *reasoning archetypes*—*linear planners* versus *reflective validata_analysists*.

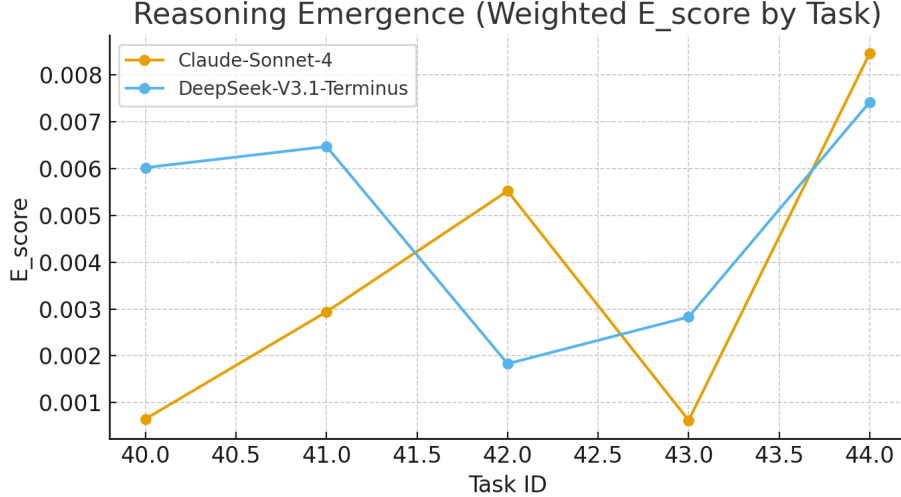


Figure 2: Reasoning emergence by task. Each point represents a data_analysista-analysis task (IDs 40–44). The **E_score** quantifies emergent reasoning strength from structural and temporal features (SDS, TCR, SCR, RD, and activity ratios). DeepSeek-V3.1-Terminus exhibits more volatile but higher peaks (notably Task 44), while Claude-Sonnet-4 shows steadier yet lower emergence across tasks.

4 Interpretation and Discussion

Across the representative data_analysista-analysis trajectories (Tasks 40–44), the two models reveal distinct reasoning styles under the GSB framework. **Claude-Sonnet-4** shows slightly higher G and stronger text compression, reflecting an early and concise articulation of objectives followed by direct, linear execution. Its responses are compact and plan-first, efficient in setting up intentions but less persistent in iterative reasoning.

In contrast, **DeepSeek** demonstrates higher B and moderately stronger S across multiple tasks. This pattern aligns with its trajectory: DeepSeek typically begins with a clear goal ($G \approx 0.30$) and then engages in self-checking and correctional steps — using phrases like “check”, “verify”, or “fix” — showing a reflective, verification-oriented process. The higher B (0.075–0.081 vs. 0.038–0.059) captures this behavior quantitatively.

Evaluation logic. Within GSB, G reflects goal clarity, S reflects structured reasoning, and B captures self-monitoring and correction. Since these tasks emphasize reliability in data_analysista

analysis, B carries greater interpretive weight. Although Claude excels in brevity and clarity of initial planning, DeepSeek’s repeated self-verification and reflective adjustments contribute more to reasoning stability.

Overall interpretation. The two models embody complementary cognitive styles:

- **Claude-Sonnet-4** → **Concise Planner:** favors linear, efficient execution with minimal redundancy.
- **DeepSeek** → **Reflective Verifier:** adopts an iterative validation loop, maintaining steadier reasoning over longer trajectories.

In scenarios requiring procedural correctness and diagnostic depth, **DeepSeek achieves a higher overall GSB score**, whereas **Claude** emphasizes communicative economy. This contrast highlights a broader trade-off between *compression-oriented* and *verification-oriented* reasoning that we see in many modern assistants.

A Metric Framework and Interpretive Logic

This appendix presents the metric definitions and the interpretive logic used throughout the paper. It formalizes how each behavioral indicator (G, S, B, SDS, TCR, SCR, RD) is computed and how these metrics map to cognitive and reasoning behaviors.

A.1 Core Lexicons

We define three regex lexicons that capture planning, verification, and debugging intents:

Π_{plan} : plan|step|strategy|outline|pipeline|procedure|first|then|next|we will

Π_{verify} : check|verify|test|assert|unit test|print|plot|compare|evaluate

Π_{fix} : fix|bug|error|patch|adjust|correct|refactor|resolve|rerun

A.2 Core Metrics (GSB)

Definition 1 (Goal Priming (G)). *Measures whether the assistant initiates planning early:*

$$G = \min(1, 0.3 + 0.7 \cdot \text{density}(\text{first_chunk}, \Pi_{\text{plan}})).$$

Definition 2 (Strategy Density (S)).

$$S = \text{density}(\text{assistant_text}, \Pi_{\text{plan}}),$$

representing sustained planning and structured reasoning.

Definition 3 (Bug/Verification Density (B)).

$$B = \max(\text{density}(\text{assistant_text}, \Pi_{\text{verify}}), \text{density}(\text{assistant_text}, \Pi_{\text{fix}})),$$

capturing self-checking and debugging tendencies.

Weighted variants. Weighted forms (G_w, S_w, B_w, RD_w) use density_w , where time-based dwell weights emphasize behaviors occurring later in the assistant trajectory. This adjustment is especially relevant for reflective behaviors (RD_w), which tend to emerge in later phases.

A.3 Auxiliary Metrics

- **SDS (Strategic Density Score):** Fraction of planning, verification, and fixing sentences within assistant outputs.
- **TCR (Textual Compression Ratio):** Ratio of unique to total sentences (higher means less redundancy and greater conciseness).
- **SCR (Sanity/Correction Ratio):** $\text{density}(\text{assistant_text}, \Pi_{\text{verify}} \cup \Pi_{\text{fix}})$, measuring the frequency of validation or correction acts.
- **RD (Reflective Density):** Proportion of reflective cues (“wait”, “rethink”, “revisit”, “oops”). A time-weighted variant RD_w captures late-phase reflection.

A.4 Cognitive and Behavioral Interpretation

Table 4 summarizes how each metric family maps to reasoning abilities and observable behavior.

Metric / Group	Cognitive Capability Captured	Observable Behavioral Pattern	Interpretive Logic
GSB (Goal–Strategy–Bug)	Macro-level task understanding, planning, and execution	Linear vs. cyclic trajectories	Indicates whether the model maintains goal alignment and structured completion over time.
Trajectory Archetype	Reasoning architecture prototype	<i>Linear Planner</i> / <i>Iterative Validator</i> / <i>Reactive Fixer</i>	Contrasts direct plan–execute with iterative self-correction styles.
Emergent Reflexivity (SCR, RD)	Self-monitoring and corrective reflex	Frequency of “wait”, “check”, “verify”-type utterances	Signals meta-reasoning and self-diagnostic awareness.
SDS (Strategic Density Score)	Density of key reasoning steps	Effective reasoning tokens / total reasoning tokens	Measures reasoning compactness and signal-to-noise ratio.
TCR (Trajectory Compression Ratio)	Compressibility of reasoning path	Accuracy retained after textual compression	Evaluates redundancy vs. conciseness in the trajectory.

Table 4: Mapping between behavioral metrics, cognitive capabilities, and interpretive logic. Weighted variants (SDS_w , SCR_w , RD_w) emphasize persistence and late-phase reflection.

B Data Preprocessing and Sentence-Level Framework

Raw conversation trajectories are stored as JSON arrays of events (**role**, **type**, timestamps, nested **message**). We standardize roles (**assistant**, **user**, **system**); timestamps are used to compute total duration and assistant-active time.

B.1 Sentence Extraction

Let $S(x)$ denote all sentences in text x . For a regex set Π :

$$\text{density}(x, \Pi) = \frac{|\{s \in S(x) : s \text{ matches } \Pi\}|}{|S(x)|}, \quad \text{density}(x, \Pi) = 0 \text{ if } |S(x)| = 0.$$

B.2 Time Weighting

Given assistant events i with dwell weight w_i , we define the time-weighted density:

$$\text{density}_w = \frac{\sum_i w_i \cdot \text{hits}_i}{\sum_i w_i \cdot \text{sentences}_i}.$$

These preprocessing steps produce the normalized sentence-level corpora (`ccbench_parsed.csv` and `ccbench_parsed_weighted.csv`) used in all subsequent analyses.